

AugmentWall: An AR Firewall Game for Concretizing Cybersecurity Concepts and Examining Student Engagement, Learning, and Motivation

Syed Ali Asif, Yiqin Cao, Sothea Veng, Kristin Chisholm, Raymond Patt (posthumous),
Teomara Rutherford, and Chien-Chung Shen
University of Delaware, asifrabi, caoemma, sothea, kschi, raypatt, teomara, cshen@udel.edu

Chrystalla Mouza
University of Illinois Urbana-Champaign, cmouza@illinois.edu

Abstract – This paper presents an Augmented Reality (AR)-infused firewall game - AugmentWall, designed to enhance middle school cybersecurity education through interactive gameplay. Developed using a participatory design approach, the game underwent iterative refinements and was playtested with 37 students across two schools to assess engagement, comprehension, and perceived educational value. This study examines how pre-game motivation influences engagement and how students perceive the game's clarity and effectiveness in teaching firewall concepts. Findings indicate that intrinsic motivation significantly impacted engagement, with highly motivated students demonstrating greater sustained participation. The game effectively reinforced firewall rule comprehension, supported by structured decision monitoring. However, prior cybersecurity knowledge was inversely related to self-reported learning gains ($r = -0.34$, $p = 0.106$), suggesting that novice learners benefited most, while experienced students may require adaptive challenges. These results highlight the potential of AR-based cybersecurity education to enhance engagement and learning while emphasizing the need for scalable difficulty adjustments to accommodate diverse learners. Future iterations should explore personalized learning pathways and extended gameplay features to optimize long-term impact.

Index Terms - Augmented Reality, Cybersecurity Education, Firewall, Interactive Learning.

INTRODUCTION AND OVERVIEW

As digital connectivity grows, early cybersecurity education is crucial to mitigate hacking, phishing, and misinformation. It builds foundational skills, fosters data protection awareness, and prepares future generations. Given that some adults struggle to adopt cybersecurity paradigms [1] and national security experts have identified workforce gaps in cybersecurity [2], early education plays a crucial role in

building foundational cybersecurity knowledge. To support this effort, we design and evaluate an interactive Augmented Reality (AR)-based firewall game to help middle school students concretize fundamental cybersecurity concepts.

Game-based approaches in education have demonstrated their ability to foster critical thinking and problem-solving skills by enabling students to engage with abstract concepts in tangible ways [3]. These align with computational thinking principles like algorithmic reasoning, decomposition, and abstraction – key to understanding firewalls and network security [4]. Using AR technology, we develop an interactive approach to concretize abstract cybersecurity principles, making them accessible and impactful for students.

Through an iterative participatory design process involving middle school educators, AugmentWall was developed and tested with 37 grades 5-8 students across two schools. This participatory approach ensured classroom alignment, integrating valuable teacher and student feedback to refine gameplay. Such approaches mirror trends in STEM education, where interactive and participatory learning fosters student ownership and empowerment, particularly in diverse educational contexts [5].

As digital security threats become increasingly sophisticated [6], there is an urgent need for both the general public and the future cybersecurity workforce to develop a strong foundational understanding of cybersecurity [7]. However, many educators lack the necessary cybersecurity knowledge to teach these concepts effectively [8]. This gap in early cybersecurity education leaves students unprepared for the digital risks they face daily and in the future [9].

AugmentWall, immerses students in problem-solving scenarios that mirror real-world cyber challenges, making cybersecurity education engaging and effective [10], [11]. The game integrates gamification and collaborative learning, similar to NxTEL, which uses AI-driven personalization and computer-supported collaborative learning (CSCL) to boost engagement [12]. Gamification elements like real-time feedback and interactive gameplay enhance participation, knowledge retention, and motivation [13]-[15]. In

cybersecurity education, gamification strengthens conceptual understanding, critical thinking, and adaptability—key skills for the digital world [10], [11].

This game helps students develop early online safety awareness while providing educators with a ready-to-use cybersecurity resource. Understanding its iterative design can support broader adoption and enhance digital literacy. As cyber threats evolve, integrating innovative, teacher-friendly learning tools into classrooms will play a crucial role in cultivating a more cyber-aware society and bridging the growing workforce gap in cybersecurity fields.

This study investigates key research questions on the effectiveness of AR-based cybersecurity education: (1) pilot students' perceptions of AugmentWall in terms of engagement, clarity, and educational value, (2) the relationship between pre-play attitudes toward cybersecurity and post-game perceptions and engagement, (3) how students interpret and apply firewall rule formation through gameplay and how the sorting map assists in tracking their understanding, and (4) the influence of baseline cybersecurity knowledge on students' understanding of firewall mechanics.

AR provides educators an immersive, practical alternative to VR, which demands more bandwidth and specialized hardware. Its accessibility makes it ideal for middle school classrooms [16]. Studies show AR enhances interactivity, content comprehension, and long-term memory retention [17]–[19]. For example, AR-based videos improve students' grasp of complex 3D spatial concepts [20]. However, effective implementation and feedback are essential, as poorly designed AR can hinder learning [21].

AugmentWall introduces firewalls as fundamental cybersecurity tools that regulate network traffic, preventing unauthorized access while allowing legitimate communication. Through interactive gameplay, students gain hands-on experience with firewall operations. Playtesting with 37 students confirmed its effectiveness in enhancing engagement and understanding.

This paper outlines AugmentWall's development, design, and playtesting, emphasizing its participatory design approach and the feedback gathered during playtesting. Future work will explore the game's impact on students' cybersecurity knowledge and expand this methodology to address additional cybersecurity concepts.

OBJECTIVES AND METHODOLOGIES

We created an AR game to educate middle school students about firewall concepts. We also wanted to observe students' motivation and attitude towards cybersecurity education. We conducted a student feedback survey to assess the game's effectiveness and enjoyment.

I. Firewalls

Firewalls protect networks by filtering data packets based on security rules, allowing or blocking access. Typically software-based, they act as a barrier between public and private networks, ensuring only trusted traffic flows through.

II. Participatory Design / Iterative Development

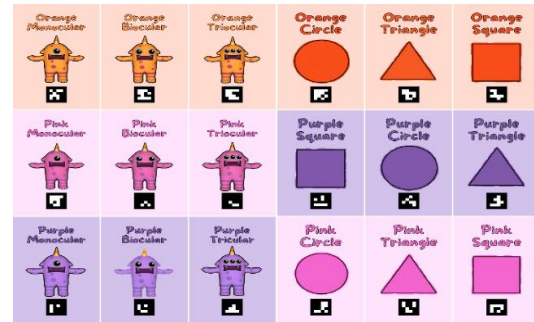


FIGURE I
PLAYING CARDS (MONSTERS AND DIFFERENT SHAPES AND COLORS)

AugmentWall was developed using an iterative design methodology [22] based on participatory design [23], ensuring continuous refinement through stakeholder feedback. Middle school teachers collaborated as experts to align the game with classroom needs. This approach, similar to heuristic teaching methods [24], emphasized collaboration and adaptability in educational tools. Teachers reviewed prototypes and storyboards during design sprints, enriching gameplay and ensuring practical implementation [25]. Table I summarizes key suggestions from the teachers.

In playtesting, students used physical cards to simulate data packets, analyzing features like shapes and colors to develop analytical reasoning collaboratively [26]. A sorting mat helped track decisions and recognize firewall rules. Our analysis examined whether students could (1) categorize packets, (2) identify patterns in firewall rule enforcement, and (3) refine reasoning about cybersecurity threats. Observing their interactions with the mat allowed us to assess its role in supporting pattern recognition and decision-making.

The game conveys three key principles: (1) firewalls block unauthorized data based on rules, (2) some data is allowed, and (3) firewalls assess both good and bad data by specific features. AugmentWall integrates these concepts to simplify cybersecurity education and enhance student engagement.

TABLE I
SUMMARY OF SUGGESTED IMPROVEMENTS FOR AUGMENTWALL

Improvement Area	Suggested Improvements
Narration Speed	Slow down the narration to make instructions easier to follow.
Completion Screen	Add a "You Won" screen to indicate successful game completion.
Game Directions	Provide clearer directions, especially for beginners.
Engagement Breaks	Incorporate a brain break video after each level to maintain engagement.
Collaboration Feature	Add a collaboration feature for players to work together, enhancing educational experience.
Walkthrough Assistance	Slow down explanations and offer more detailed walkthroughs.
Interactive Elements	Add visual aids to increase engagement and educational value.

III. Overview of the Game

The game progresses through levels of increasing complexity, following teacher-recommended cybersecurity concepts. It starts with simple shapes, then adds colors, and finally features monsters with diverse traits. Students scan cards to introduce virtual objects into the AR environment and decide whether the firewall should block them. Real-time AR feedback enhances engagement and learning.

Correct choices boost the protection score, while allowing malicious objects raises the virus score. Reaching a set protection score rewards students with AR animations, but exceeding the virus score ends the game without a reward. Teachers noted that dynamic AR rewards would enhance engagement. Student motivation influences learning outcomes, with engagement driven by personal beliefs and task value [27]. Game-like activities sustain interest and improve learning [28], [29]. These findings highlight the potential of AR-based cybersecurity education to enhance engagement and understanding. To maintain challenge, rules for shapes, colors, and monsters are randomized before each iteration, ensuring smooth classroom integration.

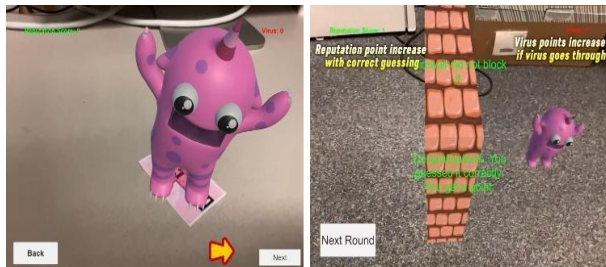


FIGURE II

AN AR OBJECT PLACEMENT ON TOP OF THE PLAYING CARD AND A SNAPSHOT OF THE AR ANIMATION PLAYED IN AUGMENTWALL

Students act as ethical hackers - cybersecurity experts who identify and fix vulnerabilities [30]. Ethical hacking involves analyzing threats, testing security, and problem-solving, making it a strong framework for cybersecurity education. Through role-based problem-solving, students analyze firewall rules, categorize data packets, and refine their reasoning. This approach fosters critical thinking, pattern recognition, and strategic decision-making.

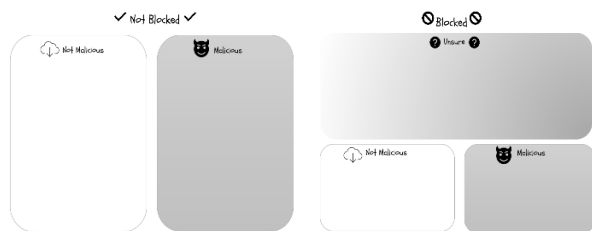


FIGURE III

SORTING MAPS USED FOR TRACKING THE MALICIOUSNESS AND BLOCK STATUS OF CARDS

Initially, we considered a two-team format where students would collaboratively design firewall rules during the design sprint. However, teacher feedback indicated that this approach could be too complex due to the additional cognitive load required for both creating and applying firewall rules, potential classroom management challenges, and the varying levels of students' prior cybersecurity knowledge. To simplify gameplay while maintaining engagement, we automated firewall rule generation, letting students focus on recognizing and analyzing patterns rather than creating rules from scratch.

IV. School-Based Playtesting and Feedback

Playtesting involved 37 students from two middle schools, requiring both parental consent and student assent. Before introducing the game, a motivation survey was administered to gauge students' baseline attitudes toward cybersecurity. This survey included questions about their general interest in cybersecurity, their understanding of its importance, and their willingness to engage with related topics. These responses provided a foundation for contextualizing their engagement with the firewall game.

After the survey, students were introduced to AugmentWall and guided through its features, including playing cards and sorting maps. They played multiple rounds, interacting with physical cards representing data packets and making decisions about which cards the firewall should block or allow. Observations were recorded through informal field notes, capturing how students categorized data packets, engaged with the sorting maps, and adjusted their strategies over multiple rounds. Notes focused on students' ability to recognize patterns in firewall rule enforcement, refine their reasoning, and navigate challenges during gameplay. Post-game feedback survey assessed their experiences, including engagement level, clarity of instructions, and perceived educational value. These qualitative observations were later reviewed alongside survey responses to examine student engagement, learning outcomes, and the clarity of game mechanics.

Table II organizes survey questions by overarching themes rather than strictly by psychological constructs such as intrinsic value or self-efficacy. Since the motivation and feedback surveys contain distinct questions rather than direct pre- and post-equivalents, the themes were derived through a systematic review and analysis of survey content to provide a structured comparison. The grouping reflects key aspects of student interaction with the game, including engagement, understanding, applicability, motivation, and confidence. By organizing the questions in this manner, the table provides a clearer way to interpret how different dimensions of student experience were assessed across both surveys without assuming a direct one-to-one correspondence between them.

The survey data will be analyzed in the results section to evaluate the game's impact and identify areas for improvement. Insights from playtesting sessions informed how students engaged with the game and its educational

effectiveness. Future iterations will integrate this feedback to refine mechanics and optimize classroom implementation.

TABLE II
THEMATIC GROUPING OF MOTIVATION AND FEEDBACK SURVEY
QUESTIONS FOR STRUCTURED COMPARISON

Theme	Motivation Survey Questions with Constructs	Feedback Survey Questions
Engagement	I like learning about cybersecurity (Intrinsic Value) [31] I enjoy dealing with the topics in cybersecurity (Intrinsic Value) [31]	The game kept me engaged.
Understanding	Knowing about cybersecurity has many benefits in my daily life (Utility for daily life) [31] I can learn a lot about cybersecurity (Self-efficacy).	I feel I better understand how firewalls work after playing the game.
Applicability	What we learn about cybersecurity is directly applicable in everyday life (Utility for daily life) [31] Knowledge of cybersecurity will be helpful for my future career (Utility for job) [31]	I can connect the concept of firewalls with the game. I would recommend this game to others for learning about firewalls.
Motivation	Cybersecurity is very important to me personally (Personal Importance) [31] I plan to contribute to my community with my cybersecurity knowledge (Social Good)	The game was enjoyable.
Skills/ Confidence	I am confident in my ability to understand cybersecurity (Self-efficacy) I believe I can master cybersecurity knowledge and skills (Self-efficacy) [32]	I was able to use the sorting mat to play/win the game.
Suggestions/ Improvements	-	10. What did you like most about the game? 11. What improvements would you suggest?

FORMULATION OF THE FIREWALL GAME

In AugmentWall, students are engaged in logical deduction and inference [33], where they must determine whether a given firewall blocks a data packet (represented by a card) and whether it is malicious. The game is structured into three levels with increasing complexity:

- **Level 1:** Cards are defined by basic geometric shapes.
- **Level 2:** Cards are defined by both geometric shapes and colors.
- **Level 3:** Cards represent monsters with various features, such as color, number of eyes, and other attributes.

I. Card Representation

Each card C in the game is represented as a vector of features:

$$C = (f_1, f_2, \dots, f_n) \quad (1)$$

Where each f_i represents a unique feature of the card. For instance, in Level 2, features might include shape (f_{shape}) and color (f_{color}).

II. Blocking Status and Maliciousness

A firewall uses two rules, R_B for blocking status and R_M for maliciousness, to determine a card's overall status. Two functions define these:

- **Block Status Function $F_{\text{block}}(C)$:**

$$F_{\text{block}}(C) = \begin{cases} 1 & \text{if } C \text{ is blocked} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

- **Maliciousness Function $F_{\text{malicious}}(C)$:**

$$F_{\text{malicious}}(C) = \begin{cases} 1 & \text{if } C \text{ is malicious} \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

A card can be blocked without being malicious or allowed despite being malicious, adding complexity to the game. Students must decipher firewall rules and blocking criteria. For example, a card may be blocked due to similarities with malicious ones but might not be inherently harmful, while another may pass through despite being potentially dangerous.

Students use the sorting mat to track each card's block status and maliciousness, helping them identify patterns in firewall decisions. As they progress, they analyze recorded data to refine their understanding of firewall logic, adjusting decisions based on trends and inconsistencies.

III. Firewall Rule Function

Each rule $r_{\{j,B\}}$ in the firewall's blocking rule set R_B and each rule $r_{\{k,M\}}$ in the maliciousness rule set R_M are logical conditions applied to the card's features. For instance, in Level 2, a blocking rule may consider different shapes:

$$r_{j,B}(C) = \begin{cases} 1 & \text{if } f_{\text{shape}} = \text{rectangle or } f_{\text{shape}} = \text{square} \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

A maliciousness rule might consider only the color:

$$r_{k,M}(C) = \begin{cases} 1 & \text{if } f_{\text{color}} = \text{Purple} \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

The overall decision functions for blocking $F_{\text{block}}(C)$ and maliciousness $F_{\text{malicious}}(C)$ can be expressed as:

$$F_{\text{block}}(C) = \bigvee_{j=1}^m r_{j,B}(C) \quad (6)$$

$$F_{\text{malicious}}(C) = \bigvee_{k=1}^n r_{k,M}(C) \quad (7)$$

Where m and n are the number of rules determining block status and maliciousness, respectively.

The combined status of card C is determined by evaluating both the blocking status and the maliciousness, resulting in the following overall decision function:

$$F_{status}(C) = \begin{cases} \text{blocked and malicious} & \text{if } F_{block}(C) = 1 \text{ and } F_{malicious}(C) = 1 \\ \text{not blocked and malicious} & \text{if } F_{block}(C) = 0 \text{ and } F_{malicious}(C) = 1 \\ \text{blocked and not malicious} & \text{if } F_{block}(C) = 1 \text{ and } F_{malicious}(C) = 0 \\ \text{not blocked and not malicious} & \text{if } F_{block}(C) = 0 \text{ and } F_{malicious}(C) = 0 \end{cases} \quad (8)$$

This function $F_{status}(C)$ categorizes each card based on its blocking status and maliciousness, comprehensively assessing how the firewall treats different cards.

IV. Logical Deduction Process

Students begin by hypothesizing which features, like shapes or colors, trigger a card's blockage or malicious classification. They test these hypotheses by selecting cards, observing outcomes, and recording results in the sorting map. This tool helps them organize findings and identify firewall behavior patterns. As students iterate, their understanding of firewall logic deepens. Broad hypotheses narrow as they collect data, improving their ability to predict how the firewall will respond. This iterative process sharpens critical thinking and logical reasoning, enabling confident predictions about which cards will be blocked or flagged as malicious.

V. A Level 2 Example

Consider the following rules in Level 2:

- The color purple is associated with maliciousness.
- The shape - rectangle is associated with being blocked.
- The shapes - triangle and circle are not blocked.

Given these rules, we describe a step-by-step example of the logical deduction process with mathematical formulations:

Step 1: Students choose a purple triangle card C_1 with features $\{f_{shape} = \text{triangle}, f_{color} = \text{purple}\}$. They hypothesize that it will be blocked. The blocking function and maliciousness function are evaluated as follows:

$$F_{block}(C_1) = r_{j, B}(C_1) = 0 \quad (\text{not blocked}) \quad (9)$$

$$F_{malicious}(C_1) = r_{k, M}(C_1) = 1 \quad (\text{malicious}) \quad (10)$$

Since the card goes through the firewall without being blocked but is deemed malicious, the virus score increases, and the protection score does not change.

Step 2: Next, they choose an orange triangle card C_2 with features $\{f_{shape} = \text{triangle}, f_{color} = \text{orange}\}$ and guess that it will not be blocked. The functions are evaluated:

$$F_{block}(C_2) = r_{j, B}(C_2) = 0 \quad (\text{not blocked}) \quad (11)$$

$$F_{malicious}(C_2) = r_{k, M}(C_2) = 0 \quad (\text{not malicious}) \quad (12)$$

The card goes through without being blocked or flagged as malicious. The protection score increases, and the virus score remains the same.

Step 3: Students then select a purple rectangle card C_3 with features $\{f_{shape} = \text{rectangle}, f_{color} = \text{purple}\}$ and guess it will be blocked. The functions are:

$$F_{block}(C_3) = r_{j, B}(C_3) = 1 \quad (\text{blocked}) \quad (13)$$

$$F_{malicious}(C_3) = r_{k, M}(C_3) = 1 \quad (\text{malicious}) \quad (14)$$

The card is blocked, and although it is malicious, the students could not observe it from the virus score because it was blocked. The protection score increases, and the virus score remains the same.

Step 4: Finally, they choose a purple circle card C_4 with features $\{f_{shape} = \text{circle}, f_{color} = \text{purple}\}$ and guess it will be not blocked due to its shape. The functions are:

$$F_{block}(C_4) = r_{j, B}(C_4) = 0 \quad (\text{not blocked}) \quad (15)$$

$$F_{malicious}(C_4) = r_{k, M}(C_4) = 1 \quad (\text{malicious}) \quad (16)$$

The card goes through the firewall without being blocked but is deemed malicious. Both the protection score and virus score increase.

From these observations, students can logically deduce that the color purple is associated with maliciousness in this round. They also infer that triangles and circles are not blocked, while rectangles are blocked. This understanding helps them refine their strategy in future rounds.

RESULTS

A comprehensive analysis examined student feedback on AugmentWall, focusing on engagement, clarity, and educational value. By integrating qualitative observations with quantitative statistical findings, we assess students' perceptions of the game experience, the relationship between their pre-game attitudes toward cybersecurity and post-game engagement, and their understanding of firewall rule formation through gameplay. Additionally, we evaluate the role of the sorting map in tracking students' comprehension and examine how baseline cybersecurity knowledge influenced their ability to grasp firewall mechanics. Key insights from pre-game motivation surveys, post-game feedback, and engagement measures provide a structured evaluation aligned with our research questions.

I. Student Engagement, Motivation, and the Impact of Pre-Game Attitudes

Evaluating engagement, clarity, and educational value in game-based learning is key to assessing effectiveness. This study also explores how pre-game cybersecurity attitudes impact post-game engagement, analyzing statistical

relationships and broader trends. By combining these perspectives, we examine student experiences and the factors influencing their engagement.

Engagement and motivation varied by school context and grade level, as depicted in Figure IV. Grade 5 students who played the game in a cybersecurity-focused after-school program reported higher engagement ($M = 4.13$, $M = 5.00$) than Grades 6–8 students who played during class ($M = 3.94$, $M = 4.26$). While pre-game motivation scores significantly differed between groups ($F = 4.49$, $p = 0.009$), post-game enjoyment ratings did not show a significant difference ($F = 2.11$, $p = 0.098$). However, post-game engagement levels were significantly higher in the after-school setting ($F = 20.08$, $p < 0.001$), suggesting that the structured, focused environment contributed to greater sustained engagement.

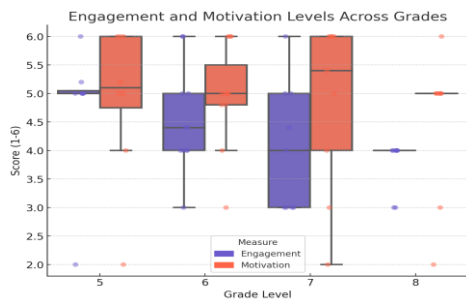


FIGURE IV

BOX-PLOTS SHOWING ENGAGEMENT LEVELS ACROSS GRADES

Further examination of how pre-game motivation influenced engagement revealed a significant positive correlation between students' intrinsic interest in cybersecurity before playing the game (Q1: "I enjoy learning about cybersecurity") and their post-game engagement levels (Q5: "The game kept me engaged") ($r = 0.29$, $p = 0.021$) illustrated in Figure V. This suggests that students with higher initial interest were more likely to remain engaged during gameplay, reinforcing the importance of motivation as a key factor in game-based learning. The average engagement score was 4.28 ($SD = 0.98$) on a 6-point Likert scale, as shown in Table III.

A K-Means clustering analysis identified three student engagement profiles shown in Figure VI: (1) highly engaged, highly motivated learners with strong pre-game motivation and sustained engagement throughout play, (2) moderately engaged, needing additional guidance who exhibits moderate motivation but requires structured scaffolding to understand game mechanics, and (3) low motivation, requires more triggers, with low initial cybersecurity motivation and reduced engagement. Students in the low-motivation cluster were particularly likely to disengage unless additional interventions were introduced, which underscored the need for pre-game activities to build interest before gameplay.

Qualitative observations revealed engagement patterns beyond statistical findings. Grade 5 students needed multiple rounds to grasp sorting mechanics but stayed engaged in collaborative play. Some felt the game ended too soon,

suggesting extended sessions could improve learning. Students with lower pre-game motivation took longer to engage, often requiring structured guidance or repeated attempts for sustained participation.

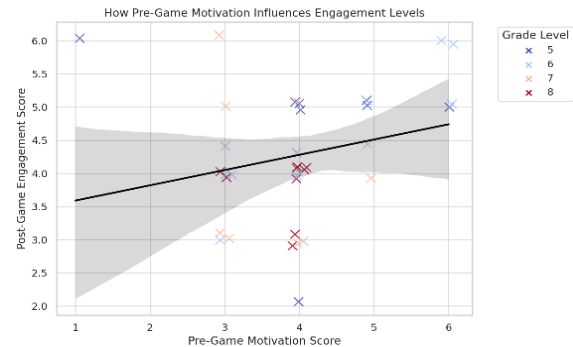


FIGURE V

A SCATTER PLOT SHOWING HOW PRE-GAME MOTIVATION INFLUENCES ENGAGEMENT LEVELS

These findings indicate that the game maintains engagement for students with intrinsic cybersecurity motivation while highlighting the need for targeted interventions to support those with lower initial interest. Future iterations may benefit from personalized learning pathways, pre-game tutorials, and adaptive difficulty features to sustain engagement across varying motivation levels. Additionally, integrating structured pre-game activities to stimulate interest and reinforce key cybersecurity concepts may help bridge the engagement gap for students with lower intrinsic motivation.

II. Student Interpretation of Firewall Rule Formation and Sorting Map Utilization

Understanding how students internalized firewall rule formation through gameplay and how the sorting map facilitated their reasoning is essential for evaluating the game's educational impact. Unlike the previous discussion on engagement and motivation, this section focuses on students' ability to analyze firewall decisions, recognize patterns, and apply logical reasoning using structured gameplay elements.

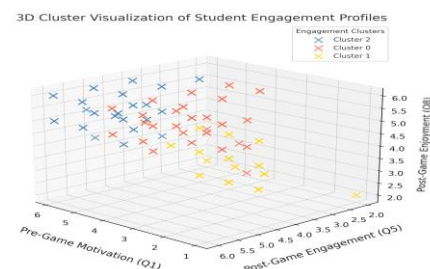


FIGURE VI

3D CLUSTER VISUALIZATION OF STUDENT ENGAGEMENT: HIGHLY ENGAGED (BLUE), MODERATELY ENGAGED (RED), AND LOW-MOTIVATION (YELLOW), HIGHLIGHTING THE NEED FOR PRE-GAME SCAFFOLDING

Survey findings indicate that students effectively connected the firewall concepts to gameplay. Their perceived understanding of firewalls was rated $M = 5.12$, $SD = 0.86$ (Q6), and their ability to relate firewalls to the game scored $M = 4.71$, $SD = 0.91$ (Q3), demonstrating that the game's interactive nature supported conceptual learning. The sorting map played a crucial role in this process, providing a structured framework for students to track firewall rules systematically.

Students rated the usability of the sorting map highly ($M = 4.84$, $SD = 0.91$, Q7), with most requiring 1-2 rounds to fully grasp its function before using it effectively. Notably, a strong positive correlation was found between students' self-rated ability to use the sorting mat and their perceived learning gains ($r = 0.32$, $p = 0.016$), suggesting that those who effectively leveraged the sorting map reported a clearer understanding of firewall mechanics. Additionally, sorting mat usage was strongly correlated with students' ability to connect firewall concepts with gameplay ($r = 0.61$, $p < 0.001$), reinforcing its importance in helping students visualize and apply rule-based cybersecurity logic.

TABLE III
DESCRIPTIVE STATISTICS SUMMARY (MOTIVATION SURVEY AND
FEEDBACK SURVEY)

Variable	Mean	Std Dev	Min	25%	Max
Pre-Game Motivation (Q1)	4.00	1.05	1.0	3.0	6.0
Post-Game Engagement (Q5)	4.28	0.98	2.0	4.0	6.0
Perceived Learning (Q6)	5.12	0.86	3.0	4.4	6.0
Sorting Map Usability (Q7)	4.84	0.91	3.0	4.0	6.0
Game Enjoyability (Q8)	4.82	1.20	2.0	4.8	6.0

The sorting map's effectiveness extended beyond comprehension to firewall rule formation—how students inferred the logic behind firewall decisions. The ease of understanding game mechanics without detailed instructions was rated $M = 4.93$, $SD = 0.90$ (Q4), and a moderate correlation ($r = 0.44$, $p = 0.006$) between sorting mat usage and intuitive rule comprehension further suggests that structured decision-tracking enhanced students' ability to process and internalize cybersecurity concepts.

Qualitative observations reinforced these findings. Most students adapted to the sorting map after an initial learning curve, though some wanted extended gameplay to explore firewall rules further. Deeper engagement with the map correlated with greater perceived learning gains, suggesting longer sessions and adaptive rule progression could enhance learning. Guided tutorials or interactive onboarding may further support students struggling with rule interpretation.

III. Baseline Cybersecurity Knowledge and Understanding of Firewall Mechanics

To examine the influence of students' baseline cybersecurity knowledge on their understanding of firewall mechanics, we

analyzed pre-game survey responses regarding students' self-rated cybersecurity knowledge. We compared them with their post-game perceptions of learning. This analysis aimed to determine whether prior familiarity with cybersecurity concepts influenced the effectiveness of the AR-based firewall game as a learning tool.

Survey results indicate a weak but negative correlation ($r = -0.34$, $p = 0.106$) between students' pre-game cybersecurity knowledge and their perceived learning gains regarding firewalls. While not statistically significant, this suggests that students who rated themselves as having higher prior knowledge of cybersecurity tended to report lower learning gains from the game. A possible explanation is a ceiling effect, where students with more advanced prior knowledge may have already understood key firewall concepts, limiting their perceived improvement.

Conversely, students with lower initial cybersecurity knowledge reported more substantial learning gains, indicating that the game effectively served its intended purpose of introducing and reinforcing fundamental firewall concepts. This aligns with observations that students who started with minimal exposure to cybersecurity engaged deeply with gameplay mechanics and relied on structured tools, such as the sorting mat, to reinforce learning.

While these findings suggest that the game successfully supports novice learners, additional scaffolding or adaptive challenge levels may benefit students with prior knowledge. Advanced gameplay modes or adjustable difficulty can keep students with a stronger foundation engaged by preventing boredom from repetitive or overly simple content.

V. Summary

Each research question explored a distinct aspect of how students engaged with and learned from AugmentWall, forming a cohesive evaluation of its effectiveness:

- **Engagement & Pre-Game Attitudes:** Higher initial motivation led to greater engagement, particularly in after-school settings.
- **Firewall Rule Formation & Sorting Map:** The sorting map improved comprehension, with a strong correlation ($r = 0.32$, $p = 0.016$) between its use and perceived learning gains.
- **Baseline Knowledge & Learning Gains:** Prior knowledge influenced perceived learning, with experienced students reporting lower improvement ($r = -0.34$, $p = 0.106$), suggesting a ceiling effect where the game was most effective for novice learners.

These findings confirm that (1) motivation drives engagement, (2) the sorting mat reinforces learning and (3) Prior knowledge shapes perceived gains, highlighting a need for adaptive challenges.

CONCLUSION

This study investigated using an AR-enhanced firewall game to introduce firewall principles to middle school students. The results show that the game effectively involves students in

decision-making learning. Notably, the role of pre-game motivation was significant in encouraging continued participation, highlighting the need for engaging and stimulating learning experiences. The sorting map was a valuable mechanism for assisting students in keeping track of and implementing firewall rules in a way that supported structured thinking. However, students with previous knowledge of cybersecurity indicated lower perceived learning gains, suggesting that the game was most helpful for students who did not know much about the subject.

Future versions should include adaptive difficulty levels to more effectively accommodate students with different amounts of prior knowledge. This commitment to inclusivity and personalized learning will enhance the game's effectiveness. Also, increasing gameplay time and incorporating structured post-game reflection can increase learning outcomes. Such improvements will ensure that AR-based cybersecurity learning continues to be accessible and effective in various learning settings.

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